

Task 1: Create a new directory called mysite in your home folder, place an index.html file inside it, and set permissions so only your user can read and access the site files.

Ans :-

Step 1: Create the website directory

```
mkdir -p ~/mysite
```

Step 2: Create an HTML page

```
echo "<h1>Welcome to My Site</h1>" > ~/mysite/index.html
```

Verify:

```
cat ~/mysite/index.html
```

Output:

```
<h1>Welcome to My Site</h1>
```

Step 3: Restrict permissions

Only your user should have access.

```
chmod 700 ~/mysite
```

```
chmod 600 ~/mysite/index.html
```

Verify:

```
ls -ld ~/mysite
```

```
ls -l ~/mysite
```

Example output:

```
drwx----- 2 student student 24 Jul 31 18:00 mysite
```

```
-rw----- 1 student student 29 Jul 31 18:00 index.html
```

Task 2: Configure your Apache or Nginx server to serve the mysite directory as a website, then test if you can access it from your browser while logged in as your user.

Ans :-

Step 1: Install Apache

```
sudo dnf install httpd -y
```

Step 2: Start and enable Apache

```
sudo systemctl enable --now httpd
```

Check status:

```
sudo systemctl status httpd
```

Step 3: Copy website files

Apache serves files from /var/www/html.

```
sudo mkdir -p /var/www/html/mysite
```

```
sudo cp ~/mysite/index.html /var/www/html/mysite/
```

Step 4: Set Apache ownership

```
sudo chown -R apache:apache /var/www/html/mysite
```

Step 5: Set permissions

```
sudo chmod 755 /var/www/html/mysite
```

```
sudo chmod 644 /var/www/html/mysite/index.html
```

Step 6: Configure SELinux

If SELinux is enforcing:

```
sudo restorecon -Rv /var/www/html
```

or

```
sudo chcon -R -t httpd_sys_content_t /var/www/html/mysite
```

Step 7: Allow HTTP through the firewall

```
sudo firewall-cmd --permanent --add-service=http
```

```
sudo firewall-cmd --reload
```

Step 8: Test in browser

Find your IP:

```
ip addr
```

or

hostname -I

Open:

http://localhost/mysite

or

http://YOUR_IP/mysite

Expected output:

Welcome to My Site

**Task 3: Change the permissions of the mysite directory so that another user on the system cannot access the site files directly. Try to access the files as the other user and note the result.

Hint: Use chmod and su or a different terminal window for the second user.**

Ans :-

Suppose another user is named **testuser**.

Switch to that user:

su - testuser

Try accessing:

cd /home/student/mysite

Expected:

bash: cd: Permission denied

Try reading the file:

cat /home/student/mysite/index.html

Expected:

cat: /home/student/mysite/index.html: Permission denied

Verify permissions:

ls -ld /home/student/mysite

Output:

drwx----- 2 student student ...

The 700 permission means:

- Owner: Read, Write, Execute
- Group: No access
- Others: No access

Therefore, other users cannot enter the directory or read the files.

Task 4: Document the steps you took to restrict access to your hosted site, including the commands used, permission settings, and screenshots or outputs showing successful and failed access attempts.

Ans :-

Commands Used

```
mkdir ~/mysite
```

```
echo "<h1>Welcome to My Site</h1>" > ~/mysite/index.html
```

```
chmod 700 ~/mysite
```

```
chmod 600 ~/mysite/index.html
```

```
sudo dnf install httpd -y
```

```
sudo systemctl enable --now httpd
```

```
sudo mkdir -p /var/www/html/mysite
```

```
sudo cp ~/mysite/index.html /var/www/html/mysite/
```

```
sudo chown -R apache:apache /var/www/html/mysite
```

```
sudo chmod 755 /var/www/html/mysite
```

```
sudo chmod 644 /var/www/html/mysite/index.html
```

```
sudo restorecon -Rv /var/www/html
```

```
sudo firewall-cmd --permanent --add-service=http
```

```
sudo firewall-cmd --reload
```

Permission Settings

Home directory:

mysite

drwx----- (700)

index.html

-rw----- (600)

Apache directory:

/var/www/html/mysite

drwxr-xr-x (755)

/var/www/html/mysite/index.html

-rw-r--r-- (644)

Successful Access

Browser:

http://localhost/mysite

Output:

Welcome to My Site

Failed Access as Another User

Command:

```
su - testuser
```

```
cat /home/student/mysite/index.html
```

Output:

Permission denied